

Where Confidence Gets Laundered: Message versus Reader in a Model Handoff

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Abstract

How much of a sender’s uncertainty survives a handoff between two frozen language models, and where is the rest lost? We measure four hand-offs from K2-Horizon-3.7B to 7B on 377 SQuAD items whose passages are clean, contradicted, or have their answer sentence removed: the passage as text, the sender’s answer with a confidence word (also as a number, and as its full candidate list with probabilities), and the sender’s KV cache through a learned per-head map. Scored as the AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, the ordering is text (.597 / .615 / .671) $\gtrsim$ mapped cache (.585 / .605 / .624) $>$ confidence word (.552 / .542 / .592); the cache-minus-word gap is .03–.06, and its paired, passage-grouped interval excludes zero on one variant of three. Two audits locate the loss. Scored directly as a label, the confidence word reaches .955 / .942 / 1.000 against the same target—it is built from that entropy, so this is a sanity check, not a result—yet this base-model receiver, under one completion prompt with no instruction to hedge, does not turn it into its own spread: in the text arm the loss is at the reader. In the cache arm, equal-capacity probes put most of the loss between the sender’s state (.739 / .706) and the mapped message (.557 / .579 at one position), with the receiver recovering part of it from the evidence (.643 on clean passages). A within-item intervention (same question, answer sentence removed) shows the receiver’s entropy moving with the sender’s through the cache (Spearman .280, same sign on 88% of items) and the text (.322, 90%), barely through the word (.146, 49%), and not through another item’s cache (.004, 47%), which shares the question and

rules out a common cause. Content controls calibrate the instrument: 63–73% on a binding test where the wrong episode’s cache is followed 64% of the time, and 53.0 F1 on SQuAD against 54.7 for text and 13.8 for the wrong passage’s cache. The message-versus-reader split is a one-line audit we recommend beside any verbal-confidence baseline.

1. Introduction

When one language model hands work to another it usually writes a message: its answer, perhaps with a confidence word. Shi et al. (2026) call the result *confidence laundering*: the receiver treats a confident-looking guess as a fact. They compared text and latent handoffs by how well the receiver’s confidence tracks the sender’s. One motivation is an edge-cloud stack that escalates from a small on-device model to a large one; we note at once that at the message size studied here a 512-token cache costs ~ 604 ms on a 1 Gbps link against 24 ms of receiver compute saved (Section 7), so this paper is a measurement, not a deployment recipe.

The measurement has a gap that no handoff study has closed. A low receiver-side score through a text handoff has two possible causes: the message does not carry the sender’s uncertainty, or the reader does not use it. Separating them costs one line of code—score the message itself against the same target that the receiver is scored against—and it changes what the comparison means. We call the first number the *message-level signal* and the second the *receiver-level signal*; their difference is the read-out gap. It is a difference of two AUROCs, not an information decomposition, and the cache has no label to score, so for the cache we use an equal-capacity probe and say so where we do.

This paper is that measurement, taken for four kinds of hand-off between the same two frozen models (K2-Horizon-3.7B $\rightarrow$ 7B) on the same 377 SQuAD items with clean, contradicted and answer-removed passages: the passage as text; the sender’s answer plus a confidence word, a number, or its full candidate list; and the sender’s KV cache through a learned per-head map. Three layers of control say where the numbers come from: content-free controls reported per vari-

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ant, including the variant where question-only wins; paired, passage-grouped bootstrap intervals on every difference; and a within-item intervention whose null arm shares the question, which is the design a cross-item correlation cannot supply. The content results—a binding test with a deranged-cache arm and a deranged-sender projector, SQuAD F1, layer-band read-out—are the instrument’s calibration and are presented as such.

The findings are narrow. The receiver’s entropy tracks the sender’s best when it reads the passage, nearly as well through the mapped cache, and worst through any summary of the sender’s belief (word, number, candidate list); the cache-minus-word gap is .03–.06 and its interval excludes zero on one variant of three. The confidence word scored directly reaches .955 / .942 / 1.000 (it is a tercile of the entropy the target thresholds, so this is a sanity check, not a discovery), yet the receiver reading it lands at .552 / .542 / .592: for this base model, under one completion prompt with no instruction to hedge, the text arm’s loss is at the reader. That sentence is a property of the label channel, not of handoffs in general. For the cache arm, equal-capacity probes locate most of the loss between the sender’s state and the mapped message, with the receiver recovering part of it from the evidence; the parsimonious reading is that evidence crosses and the receiver recomputes, and no experiment here isolates the sender’s spread crossing as such. What the within-item test licenses is that the receiver’s confidence moves because of the cache’s content, nothing more about why.

Contributions. (1) A two-level report of uncertainty transfer—message-level and receiver-level signal against one sender target—for four handoffs, with content-free controls per variant, paired passage-grouped intervals, and a within-item intervention with a common-cause null (Sections 4 and 5). (2) The label-only audit, which relocates the text arm’s loss from channel capacity to read-out for this receiver, and a corrected theory box that states what a capacity bound does and does not say about a thresholded target (Section 2). (3) Four-point equal-capacity probes (sender state, mapped message, receiver state via cache and via text) that place the cache arm’s loss, with the one-position caveat stated (Section 5). (4) The instrument’s calibration: a six-arm binding test, a deranged-sender projector, layer-band injection, the demonstration that continuation retention cannot tell a channel from a prompt, a cross-family pair and a failure pair, and the message’s cost in bytes and time (Sections 3 and 7). Shi et al. (2026) already compared answer-only, answer plus score, answer plus textual uncertainty and a latent carrier by AUROC against sender semantic entropy; we do not claim to be first to run that comparison. We claim to be the first to report both levels, with controls that survive a common-cause objection.

2. The Instrument

Models. Sender K2-Horizon-3.7B, receiver K2-Horizon-7B, both frozen, bf16, RoPE $\theta = 10^7$; they share a tokenizer and KV geometry (36 layers, 8 KV heads, head dimension 128). We capture the sender’s pre-RoPE K and raw V at every layer, map them, apply RoPE at the receiver, and insert the result as the receiver’s cache for the prefix positions. The receiver then reads only the question.

Projector. Stage 1: for each receiver layer ℓ and KV head h , a ridge regression from the concatenated $[K; V]$ of the sender (256-d per head) to the receiver’s $[K; V]$; source layers are the three sender layers whose K/V correlate best with receiver layer ℓ (“top-3”), which beat the aligned-layer map in a first pass. Stage 2: a gated SwiGLU residual (bottleneck 128, 151 M parameters) on top of the ridge output, trained with AdamW for 1,500 steps, effective batch 16 sequences of 1,024 FineWeb-Edu tokens, prefix 512, on the receiver’s cross-entropy over the continuation. Nothing in training mentions the tests.

Noma binding test. Each episode assigns one of eight single-token values to eight nonce names (“Noma is amber. Vela is indigo. . .”). Only the sender reads the facts. The receiver is asked “What colour is Vela?” and we score the eight values. Arms: *none* (question only), *text* (receiver reads the facts itself), *self* (receiver’s own cache re-injected; equals *text* to 10^{-4}), *raw* (sender cache injected without a map), *project* (mapped cache), *derange* (mapped cache from another episode with the same layout but a different value for the queried name). Content transfer = project – derange. For *derange* we also report the follow rate: how often the answer is the *other* episode’s value.

Passages. SQuAD v1.1 validation: the sender reads the passage (≤ 512 tokens), the receiver reads only the question and answers greedily (≤ 8 tokens, stopped at newline); token-F1 against the gold spans.

Uncertainty protocol. Following the knowledge-conflict and semantic-entropy protocols (Longpre et al., 2021; Xie et al., 2024; Farquhar et al., 2024), each SQuAD item gets three passages: *clean*, *contradicted* (one fluent sentence asserting a counter-answer, written by the 375B judge and kept only if DeBERTa-MNLI says it entails the counter-answer and not the gold: 587 of 941 survive), and *removed* (the gold sentence deleted). We keep the 377 items (344 distinct passages) the receiver answers correctly from clean text. The sender’s uncertainty is its semantic entropy (SE) over 10 sampled answers (mean 1.10 / 1.23 / 1.96 across the three variants, so the sender does notice); the target is SE thresholded at its median per variant. Ten samples make SE coarse: 38 / 40 / 28 distinct values per variant, and on removed passages 22% of items sit exactly at the threshold. The receiver’s uncertainty is its entropy at the first answer token.

Four handoffs are scored against the same target. Two are functions of the passage alone, $M = f(c)$, with the question unseen by the sender: *text* (the receiver reads the passage) and *mapped cache*. Two require the sender to answer the question, $M = f(c, q)$: *verbal* (the sender’s greedy answer and “confidence: high / medium / low”) and its variants with a number (the sender’s sample agreement, e.g. “0.8”) or the full candidate list (every answer cluster with its sampled frequency, e.g. “1929 (0.6); 1932 (0.3); 1935 (0.1)”). The four are therefore not one family of messages; the verbal arms have seen the question and the state arms have not, and we compare them anyway because that is the choice a system designer faces. The confidence word is an oracle label built from the sender’s *actual* SE terciles, favourable to the text channel; the sender does not verbalise its own confidence. The receiver’s prompt is a plain completion—“A smaller model that read the passage answered: X (confidence: *high*).” followed by the question—with no chat template, no few-shot examples and no instruction to hedge; we did not ablate it, and Section 9 says what that leaves open. Content-free controls: a *deranged cache* (another item’s mapped cache), a *zero cache*, a *moment-matched random cache*, and *question only*. The drop ratio is the cache path’s decrease in $P(\text{gold})$ from clean to removed divided by the text path’s: $(.279 - .130) / (.437 - .162) = .54$. It is a relative drop in confidence, not a fraction of uncertainty information retained, and the two paths start from different confidence levels.

What is being measured. Sender S reads c and holds a belief $p_S(y | c)$ over answers; its uncertainty target u_S is a function of that belief (here, SE thresholded at the median). A hand-off sends a message m and the receiver forms $p_R(y | m, q)$. We score two quantities against the same u_S : the *message-level signal*, the AUROC of a fixed function of m alone (for a confidence word, the word’s rank; for a cache, an equal-capacity probe on the mapped message), and the *receiver-level signal*, the AUROC of R ’s first-token entropy after reading m . Their difference is the read-out gap. It is a difference of two AUROCs, not an information decomposition, and the two levels for the cache use different instruments, so the gap is comparable within an arm, not across arms. *Capacity, stated correctly.* A message that is the argmax of p_S plus a k -level label is a function of p_S with at most $|Y|k$ values, so it carries at most $\log |Y| + \log k$ nats about p_S ; a state message is not so bounded, although what R recovers from it is bounded by the map and by R ’s reading. This limits information about the *distribution* p_S (which alternatives, how spread), not a detector’s AUROC against a thresholded target: a three-level label reaches AUROC .955 against the median split (Section 5). The bound is therefore not what limits the text arm here, and no measurement in this paper tests it; we keep it only to say what the earlier “log 3 nats” argument (Shi et al., 2026) does and does not imply. *Why recovering the belief would matter.* If R ’s recovered belief $\tilde{b}$ is within total variation ε of the sender’s b and one-step rewards lie in $[0, R_{\max}]$, then acting on $\tilde{b}$ instead of b loses at most $2R_{\max}\varepsilon$ in expected reward. This is the quantity a decision test should measure; this paper measures the read-out gap and the tracking of u_S , not regret.

Table 1. Binding transfer, 8 names $\times$ 8 values, $n = 1000$ episodes, chance 12.5%. All entries are percentages. “follow” is the rate at which the receiver, handed the deranged cache, answers with the other episode’s value; “content” is project – derange.

projector	none	text	raw	project	derange	follow	content
ridge top-3 (closed form)	10.6	100	14	21.7	11.4	18.2	10.3
+ residual, 1500 steps	10.6	100	14	63.0	4.6	63.7	58.4
residual, deranged sender	10.6	100	14	23.4	11.3	22.7	12.1

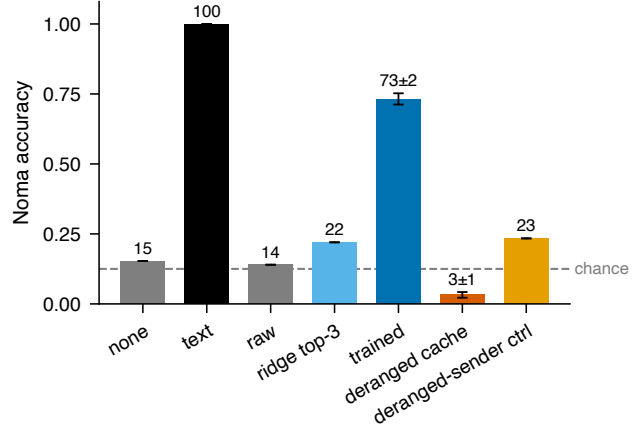


Figure 1. Noma accuracy by arm on the 6-name variant ($n = 300$). “trained” and “deranged cache” are the mean over three training seeds with error bars showing one standard deviation; the deranged-sender control is the *project* arm of the projector trained with a deranged sender ($n = 1000$, 8-name variant). Dashed line: chance (12.5%).

3. Calibrating the Instrument: Content Must Cross

Before the receiver’s confidence can track the sender’s, the cache must carry the passage. This section establishes that with the controls that the audits of Cheng et al. (2026) and Zhang & Emu (2026) ask for, and shows why the field’s usual headline number cannot.

Binding transfer. Table 1 and Figure 1. On the original 6-name variant ($n = 300$) three training seeds give 71.3 / 73.0 / 75.3% (mean 73.2, sd 2.0). Two ablations on the same variant (Appendix B): freezing the ridge and training only the residual gives 59.0%, so letting the linear map move during training is worth ~ 14 points; replacing the correlation-selected top-3 source layers with the aligned-layer map gives 32.7%, so which sender layers feed each receiver layer is the largest design choice. Matching shape alone (*raw*) does nothing. Handed the wrong episode’s cache the receiver answers with the wrong episode’s value 63.7% of the time and its accuracy falls below chance: it is not ignoring the cache, it is reading it. With the full 8-name pool and 500 episodes, the seed-2 projector gives 65.8% and 67.6% on two episode seeds (ridge: 20.4%, 20.6%). The same protocol with cities and animals gives 61.8% and

Table 2. Where the binding lives. This episode’s mapped cache is injected only into a band of receiver layers, the deranged episode’s elsewhere. $n = 500$, 6-name variant; accuracy in percent.

layers	0–5	0–11	0–17	12–23	24–35	30–35	18–35	all
accuracy	7.2	8.0	10.4	28.6	30.8	7.6	51.8	63.0

Table 3. Real passages: SQuAD v1.1 validation, $n = 300$ (600 for the trained projector). Token-F1 against the gold spans; mean gold log-probability per token.

receiver sees	none	text	raw	ridge	trained	deranged-sender ctrl	trained, derange arm
F1	16.6	54.7	0.6	14.3	53.0 (51.8)	15.1	13.8
gold log-prob / token	−3.57	−0.90	−11.2	−3.88	−1.74	−3.77	−4.18

67.2%, and padding the facts with up to 512 filler tokens does not reduce accuracy (Appendices C and E).

Deranged-sender projector. A projector trained with the identical recipe but with the sender reading text unrelated to the receiver’s continuation cannot learn a content channel, only a soft prompt. It reaches 23.4% on the binding test (Table 1, row 3). It is not content-free: its ridge base is weakly content-dependent (ridge alone 21.7%), so it controls for the *learned* part only, and we name it accordingly.

Where the binding lives. Table 2 (plotted in Appendix E): the first half of the network carries nothing usable on its own, the second half alone carries most of it, and no six-layer band suffices. The binding is read out from a distributed set of later layers; the filler used to complete the cache in these runs is the deranged episode’s cache, a plausible but wrong state.

Real passages. Table 3 and Figure 2 apply the same projectors, never trained on questions, to SQuAD. On F1 the trained projector recovers 96% of the gap between question-only and reading the passage (53.0 at $n = 300$; 51.8 at $n = 600$); on gold log-probability it recovers 69%. The wrong passage’s cache (13.8), the ridge map (14.3) and the deranged-sender projector (15.1) all sit at or below question-only (16.6). On passages, as on the binding test, the channel carries content, not a bias toward answering.

Continuation retention cannot tell a channel from a prompt. Following the closed-form-map literature (Heo et al., 2026) we also report retention (Table 4; plotted in Appendix H). Theirs is an accuracy ratio (transferred accuracy over the receiver’s own); ours is the fraction of the continuation-loss gap recovered on 256 fresh held-out sequences deduplicated against training text (1.19 for the trained projector, −0.38 for the ridge; 128 sequences in the table), so the table critiques our own metric, not theirs. The deranged-sender projector—a learned soft prompt on a weakly content-dependent ridge base—recovers 77% of the oracle gap, and almost all of it (0.72) survives when its input is deranged at test time. The trained projector’s

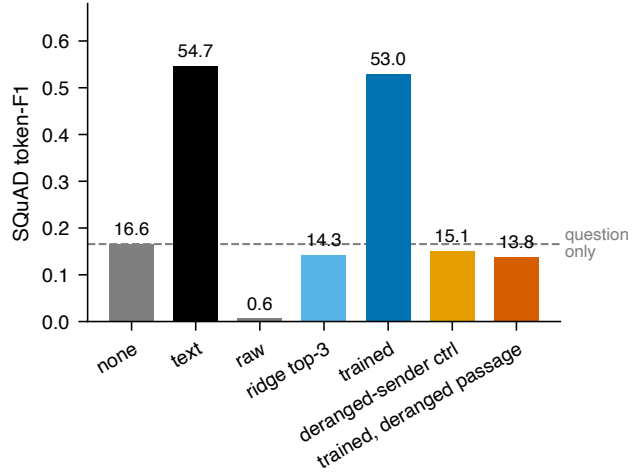


Figure 2. SQuAD token-F1 by arm ($n = 300$). Dashed line: question only. The trained projector sits within two points of the receiver reading the passage itself; the ridge map, the deranged-sender control, and the trained projector given the wrong passage’s cache all sit at or below question-only.

Table 4. Continuation retention is the wrong headline. Retention = $(\text{loss}_{\text{none}} - \text{loss}_{\text{project}}) / (\text{loss}_{\text{none}} - \text{loss}_{\text{oracle}})$ on 128 fresh held-out sequences; the second column swaps the sender’s text at test time.

projector	retention	retention, sender text deranged
ridge top-3	−0.30	−0.66
+ residual	1.23	0.31
residual, deranged sender	0.77	0.72

1.23 decomposes into ~ 0.3 of the same prompt-like effect and ~ 0.9 that disappears when the sender’s text is swapped. Retention above 1 is not evidence of communication; the binding test with its derange arm is. With that established, the instrument is calibrated: the cache carries the passage, and any uncertainty signal measured through it is measured through a channel that could in principle carry one.

4. How Much Uncertainty Survives Each Handoff

Result. Table 5 and Figure 3 (left). The ordering of receiver-level signal is the same on every variant: the receiver reading the passage, then the mapped cache, then any summary of the sender’s belief. The cache beats the verbal handoff by $+0.03$ to $+0.06$ AUROC, its drop ratio is .54 where the verbal handoff’s is .30, and it answers better on clean and contradicted passages (F1 59.8 / 49.0 vs 55.0 / 37.7). Writing the confidence as a number does not help the text channel (.548 / .532 / .563; drop ratio .32), and neither does the strongest text message we can build from the sender’s samples, every answer clus-

Table 5. Sender uncertainty through each handoff; 377 SQuAD items the receiver answers correctly from clean text. AUROC is of the receiver’s first-token entropy against the sender’s semantic entropy (thresholded at its median per variant); the drop ratio is the receiver’s $P(\text{gold})$ loss from clean to removed relative to the text path’s. The two “scored directly” rows are the message-level signal: the label itself ranked against the same target, with no receiver; they are near-tautological (the word is a tercile of the entropy the target thresholds) and serve as the audit of Section 5, not as a result. Two projectors trained with uncertainty objectives are in Appendix A.

receiver sees	$P(\text{gold})$	entropy	AUROC vs sender SE	drop ratio
	clean / contra. / removed	clean / removed	clean / contra. / removed	
text	.437 / .290 / .162	1.14 / 3.03	.597 / .615 / .671	1 (def.)
text + answer + confidence word (verbal)	.253 / .193 / .171	2.88 / 2.80	.552 / .542 / .592	.30
text + answer + numeric confidence (sample agreement, e.g. “0.8”)	.258 / .197 / .170	2.93 / 2.93	.548 / .532 / .563	.32
text + all candidate answers with their sampled probabilities	.264 / .210 / .144	2.68 / 2.63	.495 / .489 / .626	.44
mapped cache	.279 / .249 / .130	1.78 / 3.16	.585 / .605 / .624	.54
deranged cache	.105 / .106 / .105	3.88 / 3.88	.536 / .488 / .564	0
zero cache	.034 / .033 / .037	6.41 / 6.18	.496 / .481 / .502	0
moment-matched random cache	.005 / .004 / .005	7.24 / 7.30	.457 / .472 / .580	0
question only	.127	3.69	.532 / .479 / .603	0
<i>message-level signal (no receiver)</i>				
confidence word, scored directly as a label	—	—	.955 / .942 / 1.000	—
numeric confidence, scored directly	—	—	.870 / .848 / .735	—

ter with its frequency (.495 / .489 / .626; drop ratio .44; F1 55.4 / 40.7 / 22.3): the receiver’s confidence tracks the sender no better from it than from a word, except on removed passages, where it matches the cache. With paired bootstrap intervals over passages (344 passages, 2,000 re-samples), cache—verbal is significant on contradicted passages ([.008, .117]) and not on clean ([−.027, .094]) or removed ([−.024, .086]); cache—question-only is significant on contradicted ([.065, .180]) and borderline on clean ([−.005, .111]); on removed passages the text path is significantly above the cache ([−.091, −.002]). The direction is consistent; the size is modest and established on one variant of three.

Controls, per variant. The content-free controls do not sit at one level. On clean and contradicted passages they lie at .46–.54 (question only .532 / .479; deranged .536 / .488; random .457 / .472; zero .496 / .481) against the cache’s .585 / .605. On removed passages question-only alone gives .603, the random cache .580 and the deranged cache .564 against the cache’s .624: the question already carries most of the signal when the answer is gone, and question-only there beats the verbal arm (.592). The cache’s increment over question-only is therefore +.053 on clean, +.126 on contradicted and +.021 on removed passages, and only the middle one is established. When the answer is removed, the cache path loses 54% of the confidence the text path loses and its entropy rises by 73% of the text path’s rise. Under a contradiction the text path is swayed slightly more by the inserted sentence than by the original ($P(\text{counter})$.32 > $P(\text{gold})$.29), while the cache path keeps the original ahead (.25 > .22) and its F1 is higher (49.0 vs 45.2); across all conditions the cache path is less confident than the text path on clean passages ($P(\text{gold})$.28 vs .44), so a receiver that treats the two channels as equivalent would under-trust the cache. What this section cannot say is why the ordering is

what it is; that is the next section’s job.

5. Where the Loss Occurs

(a) The label carries the signal; this receiver does not read it. The confidence word is built from the sender’s own SE terciles, so scored as a label it predicts the binary target almost perfectly without any receiver: AUROC .955 / .942 / 1.000 on the three variants, the numeric label .870 / .848 / .735 (Table 5, bottom). These numbers are a property of the construction (a tercile split against a median split of the same quantity), and the 1.000 on removed passages is a degeneracy—22% of items tie at the threshold and the tercile boundary coincides with it—so the audit is a sanity check, not a finding. The finding is the other bar: the receiver’s first-token entropy after reading that word reaches .552 / .542 / .592. The text channel is not capacity-limited here; this base-model receiver, under one completion prompt, does not turn a label into its own spread. That is a statement about one reader and one prompt format; a receiver instructed to hedge, given few-shot examples, or instruction-tuned could erase the contrast, and we did not test one. It is also the reason the “log 3 nats” argument in the box of Section 2 cannot be the mechanism of the verbal arm’s shortfall.

(b) Within-item intervention. Cross-item correlations admit a common cause: a hard question makes both models uncertain whether or not anything crosses. The variants allow a paired test with the question held fixed and only the sender’s passage changed (clean → removed; the sender’s SE rises by 0.98 on average). Per item, the receiver’s entropy change tracks the sender’s SE change through the mapped cache (Spearman .280; same sign on 88% of items) about as well as when the receiver reads the text (.322; 90%), weakly through the confidence word (.146; 49%), and not at all through a cache from another item (.004; 47%), which

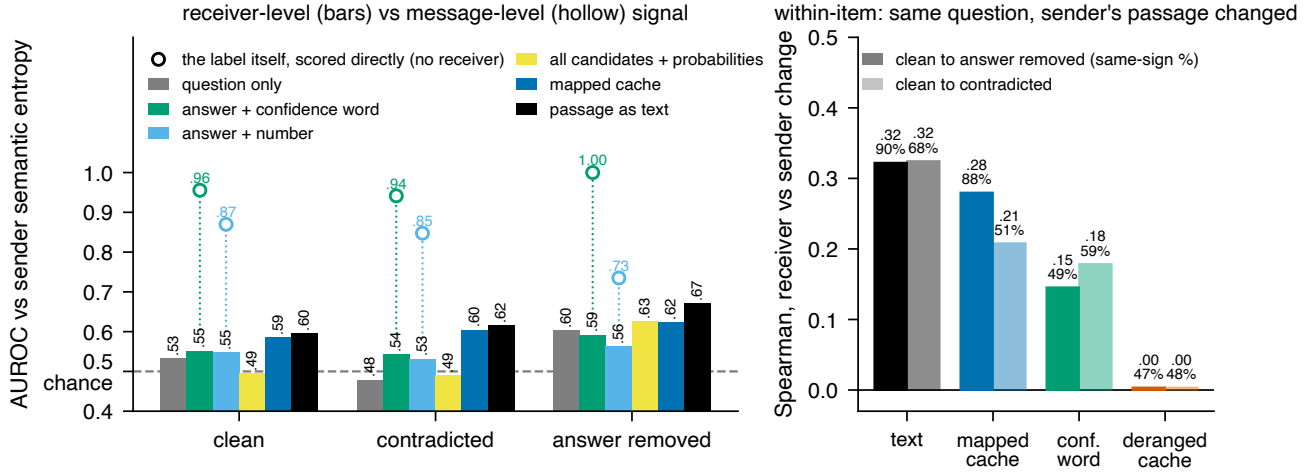


Figure 3. Left: receiver-level signal (bars: AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, 377 items) for six handoffs per passage variant, with the message-level signal of the label itself drawn as a hollow marker above the word and number arms. The vertical distance from marker to bar is the read-out gap; the marker is near 1 by construction (the word is a tercile of the same entropy), so the informative half is the bar. The cache has no label and no marker. Paired, passage-grouped intervals on the differences are given in the text. Right: within-item intervention, the question held fixed and the sender’s passage changed; bars are the Spearman correlation between the receiver’s entropy change and the sender’s semantic-entropy change, annotated with the same-sign rate.

shares the question and rules out the common cause (Figure 3, right). Under a contradiction the effect is smaller (cache .208, text .325, word .179, deranged .004). The receiver’s confidence moves with what the sender saw, and it moves because of the cache’s content. It licenses no more than that: the text path tracks at least as well, so the parsimonious reading is that the evidence crosses and the receiver recomputes its own uncertainty from it.

(c) Shape of the belief. Sender SE and receiver first-token entropy are different measurements; a diagnostic closer to the belief itself is the *shape*: per item with at least two sampled answer clusters, the Spearman correlation between the sender’s cluster frequencies and the receiver’s first-token mass on those candidates. On clean passages ($n = 323$) it is .199 through the mapped cache, .193 from the text, .166 from the explicit candidate list, and .052 / .042 for the deranged cache and question only; .153 / .234 / .201 under a contradiction ($n = 351$); near zero for every arm on removed passages. The receiver recovers the sender’s alternatives about as well from the cache as from the passage, and no better from a list of them, but these correlations are small enough that the diagnostic separates arms from the controls and not from each other.

(d) Four-point probes. Equal-capacity logistic probes for the binary sender-SE target, nested five-fold and grouped by passage on the 587 items, at four points of the chain (clean / removed): the sender’s own last-prompt-token state gives AUROC .739 / .706; the mapped message at its last prefix position, read alone, .557 / .579 (with the question-only receiver state concatenated, .563 / .603); the receiver’s state after reading through the mapped cache .643 / .563; after

reading the text .650 / .654; the question alone .530 / .591. Three things follow. The sender holds more about its own uncertainty than any receiver state recovers. On the cache path most of the loss is between the sender’s state and the mapped message, not at the reader, and the receiver on clean passages recovers more (.643) than the probed message carries (.557): the reader adds signal from the evidence. And on clean passages the cache leaves the receiver with as much as the text does (.643 vs .650), while on removed passages it does not (.563 vs .654). Two caveats bound this. The message-level probe sees one position of 512, so .557 / .579 bounds what one position carries, not what the map preserves; a pooled read-out of the whole prefix is the obvious next probe. And these three levels are three instruments—a label’s rank, a linear probe, an entropy read-out—so their differences are not a decomposition; taken as one they would put the cache arm’s loss at the reader on the text side and *before* the reader on the cache side, which is the shape of the data, not an accounting identity.

6. Two Objectives That Did Not Move the Loss

If the cache arm’s shortfall were an artefact of the projector’s training target, changing the target should move it. Two receiver-side objectives did not (Appendix A): a KL term to the receiver’s own text-path distribution leaves the drop ratio at .56 (vs .54) and AUROC within .01 while costing 2 F1, and matching the receiver’s per-token entropy through the cache to the sender’s own leaves it at .51 and costs 8 F1. Neither target contains the sender’s spread; a sender-side reader $r(m, q)$ trained on the sender’s candidate distribution and kept separate from the receiver’s own decision is the

Table 6. What the message costs. Bytes per prefix token of the mapped cache (512-token SQuAD passages; “KB” = 1024 bytes). Noma: 6-name variant, $n = 300$, seed-2 projector. “filler” fills the untransmitted layers or heads with the receiver’s own cache of a neutral text of the same length. The full sweep, including zero and mean-vector fills and smaller projectors, is in Appendix F.

message	KB / token	Noma	SQuAD F1
full mapped cache, bf16	144	75.3	53.0
int4 fake-quantised	36	75.0	51.6
layers 18–35 only, filler elsewhere	72	28.7	16.1
4 of 8 heads, filler elsewhere	72	26.7	12.8

untested fix.

7. What the Channel Costs and Where It Works

Cost. The full mapped cache is $36 \text{ layers} \times 8 \text{ heads} \times 2 \times 128 \times \text{bf16} = 144 \text{ KB}$ per token, 72 MiB for a 512-token prefix, against $\sim 1 \text{ KB}$ of text; the projector has 151 M parameters. Table 6: int8 and int4 fake-quantisation keep binding accuracy (76.0 / 75.0 vs 75.3%) and SQuAD F1 (52.1 / 51.6 vs 53.0), bytes $\div 4$, but this is fake quantisation, not a packed transport. Nothing else is free. Dropping layers or heads with zeros in their place collapses transfer (layers 18–35: 24.3%; 4 heads: 25.7%), but zeros also dilute the softmax (each zero key adds e^0 to the denominator), so two structural null states test whether the collapse is missing content or broken attention: a one-vector position-mean fill gives 33.7% for layers 18–35 and 35.0% for 12–35 (F1 21–29), and the filler null—the receiver’s own cache of a neutral text, a valid episode-free state—gives 28.7% and 30.7% (F1 16–19; 4 heads: 26.7%). Neither recovers the transfer, so with this projector the message cannot be cut by layers; what is missing is content. Low-rank truncation keeps the binding test (rank 32 / 16: 75.0%) but not passages (F1 45.3 / 29.6). Latency on one A100: the receiver re-reading 512 tokens takes 48 ms, the projector 24 ms. But at 72 MiB per message a 1 Gbps link needs $\sim 604 \text{ ms}$ (int4: $\sim 151 \text{ ms}$) against 24 ms of receiver compute saved, so the compute advantage does not carry over to a network handoff at this size. A reader who can get the sender’s uncertainty by re-reading 1 KB of text should; the channel is worth its bytes only where the sender’s state, not its passage, is the point, and this paper has not exhibited such a case.

Where it works. The recipe crosses one family boundary and fails on one same-family pair (Appendix C). With Qwen3-4B as the sender (matched KV geometry, different vocabulary and pre-training; receiver positions aligned by character offsets) the trained projector reaches 76.7 / 70.3 / 68.0% over three training seeds (mean 71.7, sd 4.5; deranged $\leq 4.7\%$, follow rate 80% for seed 0) and SQuAD F1 48.0 / 51.9 / 44.9 (mean 48.3) against 54.7 for text, while

the ridge map alone gives 18.0%: cross-family transfer is as strong as within-family on the binding test and within 7 F1 on passages. With the roles reversed (7B sender, 3.7B receiver) the pattern repeats at 63.3%. For K2-Horizon-0.9B, which differs from the receiver in vocabulary, layer count and head dimension (64) at once, the trained projector reaches 17.0% (deranged 14.0%, follow 11.3%) and 24.2 F1; this parameterisation fails on that pair, and which difference matters is untested. On HotpotQA bridge questions, where sender and receiver hold different paragraphs (Appendix D), the receiver’s F1 goes from 30.1 with its own paragraph to 39.9 ± 2.1 with the sender’s paragraph as a mapped cache, against 45.8 as text and 29.9–32.1 for a deranged cache. A handoff-policy experiment on SQuAD (Appendix G) is negative for a reason specific to the task and is summarised in Section 9.

8. Related Work

Confidence Laundering (Shi et al., 2026) is the position this paper measures: it compared answer-only, answer plus score, answer plus textual uncertainty and a latent carrier by AUROC against sender semantic entropy and found the latent carrier ahead; it did not score the message itself, report a content-free control, or intervene within an item, and its numbers are not extractable from the published version. Two causal audits supply the controls we adopt: Cheng et al. (2026) find that deranged caches lose only 0.4 points in relayed-cache setups, the objection our derange and deranged-sender arms answer, and Zhang & Emu (2026) decompose a latent channel’s effect into “sender content” and “any message present” with boundary-replacement controls, of which our deranged-sender projector isolates the second term. The channel literature around us has no message-versus-reader split: cache-to-cache communication (C2C; Fu et al., 2025) and Latent Cache Flow (Rossi et al., 2026) train fusers between models of similar size; a closed-form KV map (Heo et al., 2026) reports 44–98% continuation retention across sizes; K-V cache alignment (Dery et al., 2026) learns a map between two models’ caches, the closest setup to ours; a universal context-reuse layer (Li et al., 2026) makes one model’s KV state consumable across scale, tokenizer and family without a content-free control or an uncertainty test; StateBridge (Peng et al., 2026) aligns final-layer hidden states into the receiver’s input space with a closed-form orthogonal map; Interlat (Du et al., 2025) has agents communicate entirely in latent space and argues latents keep several hypotheses; Dense Latent Communication (Chen et al., 2026) sends latent messages across heterogeneous agents and separates context-aware from context-unaware receivers; Dual-Cache (Liu et al., 2026a), CacheBridge (Qu et al., 2026), HyLaT (Mou et al., 2026) and LatentPort (Villani, 2026) extend cache transfer to heterogeneous pairs, hybrid protocols and recurrent state;

CIPHER (Pham et al., 2023) had agents debate through embeddings. MentorPulse (Liu et al., 2026b) finds that a static one-shot latent handoff can fall below no guidance on long-form generation, a scope limit our short-answer setting does not probe; the alignment ladder of Marincat (2026) runs a leakage-controlled audit close in spirit to our derange arms; the Replay Gap (Gonuguntla, 2026) asks for branched rollouts, which remain a next step. On the uncertainty side we borrow the knowledge-conflict construction of Longpre et al. (2021) and Xie et al. (2024), semantic entropy as the sender’s uncertainty (Farquhar et al., 2024), and the finding that verbalised confidence is poorly calibrated (Tian et al., 2023; Yona et al., 2024); distillation-style maps have been observed to contract predictive variance (Cui & Pei, 2026), which we cite as an observation to compare against, not as a mechanism for our map. These works are concurrent with ours; we treat them as the surrounding conversation, not as prior results this paper improves on.

9. Limitations

The uncertainty measurement rests on 377 items of one dataset, one sender–receiver pair, one projector seed and one sender-uncertainty measure; the three projector seeds trained for the binding test have not been run through the confidence protocol, which is the cheapest missing control. The contested effect is small: cache minus verbal is .03–.06 AUROC with the interval excluding zero on one variant of three, and on removed passages question-only beats the verbal arm and text beats the cache. The absolute AUROCs are ~ 0.6 for every channel including text, so every claim is relative. “The receiver does not read the label” is a property of one base model under one completion prompt: no prompt ablation, no few-shot examples, no chat template, no instruction-tuned receiver, and the candidate-list arm scoring below question-only on clean passages (.495 vs .532) says the text baselines are not the strongest a competent engineer could build. The label-only audit is near-tautological by construction; its information is in the receiver’s bar, not the label’s. The four-point probes read one position of the mapped message and use three different instruments, so they locate the loss coarsely. The belief-shape correlations (.199 / .193 / .166) separate arms from controls, not from each other. Three controls are weaker than their names: the deranged-sender projector is not content-free (23.4% against a ridge base of 21.7%), the retention critique is of our own metric rather than of Heo et al. (2026), and the layer-band experiment uses the deranged episode’s cache as filler. The setting is context reuse—the sender’s cache of a passage prefix—not a mid-task handoff of an agent’s reasoning or tool state, and nothing here says whether any of it survives a sender that has reasoned or generated at length (Liu et al., 2026b). The channel is expensive in bytes (144 KB per token; 604 ms per message on a 1 Gbps link)

and the handoff-policy experiment (Appendix G) is negative on SQuAD because the receiver reading the passage is only 3.5 F1 above the sender (55.7 vs 52.2), so no trigger, not even a .749-AUROC probe on the sender’s hidden state, makes a handoff pay; the deployment that motivates the paper does not exist at this message size.

10. Conclusion

Score the message, then score the reader. Done for four handoffs between two frozen models, that pair of numbers turns “a text handoff launders the sender’s uncertainty” into something narrower and testable: for this receiver, a confidence word carries the signal (.955 / .942 / 1.000) and the reader discards it (.552 / .542 / .592), while a mapped KV cache with no label at all leaves the receiver tracking the sender nearly as well as reading the passage (.585 / .605 / .624 vs .597 / .615 / .671), moves its confidence within an item because of the cache’s content (.280 vs .004 for another item’s cache), and loses most of what it loses before the reader (sender state .739 / .706, mapped message .557 / .579 at one position). The measured ordering is text $\gtrsim$ cache $>$ label $>$ deranged; the cache-minus-label gap is small and established on one variant of three; and the honest reading of the cache arm is that evidence crosses and the receiver recomputes. What would overturn the paper’s one general sentence is a receiver that reads the label—an instruction-tuned or few-shot reader whose entropy follows “confidence: low”—and that is the first experiment we would run. What would make the channel worth its bytes is a case where the sender’s state, not its passage, is the point; we have not exhibited one.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here. One consideration specific to cache handoff: a cache that a receiver “believes” (Section 3) is also a cache that can be tampered with; deployments that move caches between devices should authenticate them as they would any other input.

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A. Training Objectives That Did Not Recover the Sender’s Spread

Table 7. Two projectors trained with an uncertainty objective (weight 1, 1,000 steps), scored exactly as in Table 5.

receiver sees	$P(\text{gold})$	entropy	AUROC vs sender SE	drop ratio
	clean / contra. / removed	clean / removed	clean / contra. / removed	
mapped cache (reference, Table 5)	.279 / .249 / .130	1.78 / 3.16	.585 / .605 / .624	.54
mapped cache, KL-trained projector	.272 / .241 / .118	2.38 / 3.76	.590 / .589 / .611	.56
mapped cache, sender-entropy-matched projector	.247 / .222 / .107	2.23 / 3.64	.596 / .587 / .627	.51

Adding $\text{KL}(\text{text-path} \parallel \text{cache-path})$ on the receiver’s next-token distributions to the projector’s objective (weight 1, 1,000 steps) raises the cache path’s entropy everywhere (clean 2.38 vs 1.78) without making it track the sender better (AUROC .590 / .589 / .611; drop ratio .56 vs .54) and costs 2 F1. Matching the receiver’s own text-path distribution is the wrong target: what is lost is the *sender’s* spread, which the receiver’s text path does not contain either. A sender-side target—matching the receiver’s per-token entropy through the cache to the sender’s own per-token entropy on the training text—was the second attempt (weight 1, 1,000 steps): binding accuracy holds (72.3%; Appendix B), SQuAD F1 falls to 45.3, and the uncertainty measures do not move (AUROC .596 / .587 / .627; drop ratio .51). Entropy matching is not belief matching; the untested fix is a separate reader $r(m, q)$ trained on the sender’s candidate distribution.

B. Ablations on the 6-Name Variant

All runs: sender K2-Horizon-3.7B, receiver K2-Horizon-7B, colour attribute, 6 names, $n = 300$ episodes, chance 12.5%. Accuracy in percent. The two 1,000-step rows with an added uncertainty term (KL, sender-entropy matching) use the full 151 M-parameter projector.

Table 8. Projector ablations on the 6-name Noma variant.

projector	project	derange	follow
ridge top-3 (closed form)	22.0	10.7	30.3
+ residual, 300 steps	56.3	6.3	53.7
+ residual, 1500 steps, seed 0	71.3	4.3	71.7
+ residual, 1500 steps, seed 1	73.0	3.0	74.0
+ residual, 1500 steps, seed 2	75.3	2.3	76.0
+ residual, 1500 steps, ridge frozen	59.0	7.0	59.3
+ residual, 1500 steps, aligned-layer ridge	32.7	8.0	34.7
+ residual, 1000 steps, bottleneck 32	64.3	5.7	62.3
+ residual, 1000 steps, bottleneck 16	61.3	7.7	60.0
+ residual, 1000 steps, + $\text{KL}(\text{text} \parallel \text{cache})$	66.0	5.3	63.3
+ residual, 1000 steps, + sender-entropy matching	72.3	3.0	73.3

C. Other Attributes, Reversed Roles, and Other Pairs

The channel is not one-way: with the roles swapped (7B sender, 3.7B receiver) the ridge map gives 29.0% and the trained projector 63.3%, with the deranged cache at 5.7% and a 63% follow rate—the same pattern at slightly lower accuracy. With Qwen3-4B as the sender (36 layers, 8 KV heads, head dimension 128—matched KV geometry—but a different vocabulary and pre-training), receiver positions are aligned to sender tokens by character offsets and the same recipe is run: top-3 ridge 18.0% (raw injection 15.7%, deranged 13.0%), trained projector 76.7 / 70.3 / 68.0% over three training seeds (mean 71.7, sd 4.5; deranged $\leq 4.7\%$, follow rate 80% for seed 0), SQuAD F1 48.0 / 51.9 / 44.9 (mean 48.3) against 54.7 for the receiver reading the text. The linear map alone does nothing across families. The recipe fails for K2-Horizon-0.9B, a same-family sender with a different vocabulary and head dimension 64: the trained projector reaches 17.0% (deranged 14.0%, follow 11.3%; content transfer 3 points) and 24.2 F1 on SQuAD. K2-0.9B differs from the receiver in vocabulary, layer count and head dimension at once; which difference matters is untested, and cross-head maps exist in other work, so the failure is of this parameterisation, not of the pair.

Table 9. Binding transfer with city and animal attributes ($n = 500$, 8 names) and with the roles reversed (7B sender, 3.7B receiver; 6 names, $n = 300$).

setting	projector	none	project	derange	follow
city	ridge top-3	14.2	19.8	11.8	18.4
city	trained	14.2	61.8	6.4	60.2
city	deranged-sender ctrl	14.2	15.4	11.4	18.6
animal	ridge top-3	12.2	12.8	12.0	11.2
animal	trained	12.2	67.2	5.2	64.6
animal	deranged-sender ctrl	12.2	12.4	12.8	9.8
reversed	ridge top-3	11.7	29.0	12.3	31.3
reversed	trained	11.7	63.3	5.7	63.0

Table 10. Four sender–receiver pairs, same recipe. Noma 6-name variant, $n = 300$, chance 12.5%; percentages. “proj.” is the *project* arm, “der.” the *derange* arm, “follow” the follow rate. SQuAD F1 is for the trained projector ($n = 300$; text ceiling 54.7). K2-Horizon unless named. Qwen3-4B row: proj. and F1 are mean $\pm$ sd over three training seeds (76.7 / 70.3 / 68.0; F1 48.0 / 51.9 / 44.9); der. is the worst seed, follow is seed 0. The raw arm is undefined for K2-0.9B (head dimension 64).

sender $\rightarrow$ receiver	raw	ridge top-3			+ trained residual			F1
		proj.	der.	follow	proj.	der.	follow	
3.7B $\rightarrow$ 7B	14.0	22.0	10.7	30.3	71.3	4.3	71.7	53.0
7B $\rightarrow$ 3.7B	11.7	29.0	12.3	31.3	63.3	5.7	63.0	—
Qwen3-4B $\rightarrow$ 7B	15.7	18.0	13.0	17.0	71.7 $\pm$ 4.5	≤ 4.7	80.3	48.3 $\pm$ 3.5
0.9B $\rightarrow$ 7B	n/a	13.3	13.0	10.7	17.0	14.0	11.3	24.2

D. Different Contexts: HotpotQA

Table 11. HotpotQA bridge questions, $n = 400$. Sender and receiver each hold one of the two supporting paragraphs; the receiver always reads its own paragraph and the question. Token-F1; the mapped cache is mean $\pm$ sd over three training seeds (37.9 / 39.7 / 42.1), the deranged cache their range.

receiver additionally gets	F1
nothing (own paragraph + question)	30.1
the sender’s paragraph as text	45.8
the sender’s paragraph as a mapped cache	39.9 $\pm$ 2.1
a deranged mapped cache	29.9–32.1
the sender’s paragraph through the ridge map	26.3
a zero cache	20.9

This is the LCF-X setting of Rossi et al. (2026). The receiver with its own paragraph and the question reaches F1 30.1; with both paragraphs as text 45.8; with its own paragraph plus the sender’s mapped cache 37.9 / 39.7 / 42.1 over the three training seeds (mean 39.9, sd 2.1)—63% of the gap—while the deranged cache gives 29.9–32.1 (no help), the ridge map 26.3 and a zero cache 20.9. The channel carries the *other* paragraph, and only the trained map does.

E. Distance Sweep and Layer Bands

F. Compression Sweep

A projector with bottleneck 32 has 70 M parameters and keeps SQuAD F1 (53.7) while losing 9 points on the binding test (64.3%); bottleneck 16 (62 M) gives 61.3% and 51.6 F1. The filler rows are the structural null of Section 7: the receiver’s own cache of a neutral text is a valid, episode-free state, so a collapse under it is missing content rather than broken attention.

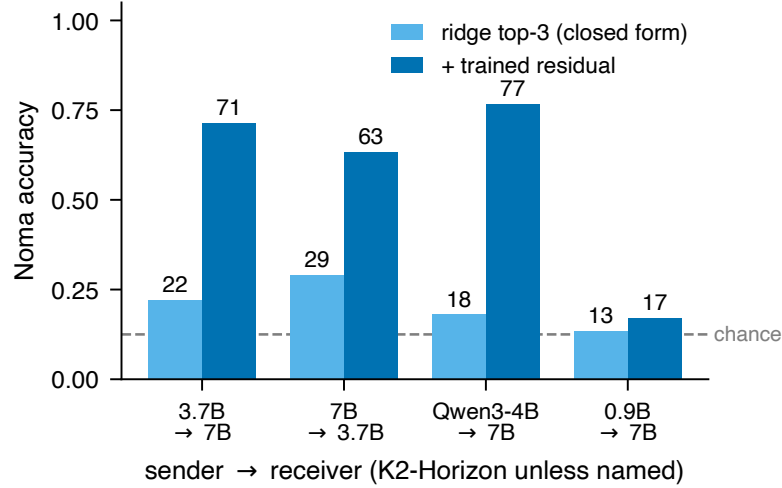


Figure 4. Noma accuracy (*project* arm) for the closed-form ridge map and the trained projector on four sender–receiver pairs ($n = 300$, 6-name variant). Dashed line: chance. The trained map crosses families (Qwen3-4B) but not a vocabulary and head-dimension change within the family (K2-0.9B).

Table 12. Noma accuracy (*project* arm) versus filler tokens between facts and question. 8-name variant; $n = 1000$ at pad 0, $n = 500$ otherwise.

projector	0	64	128	256	512
ridge top-3	21.7	22.2	22.4	22.8	24.2
trained	63.0	61.0	57.0	62.4	68.8
deranged-sender ctrl	23.4	25.0	19.8	18.2	13.8

G. When Should the Small Model Hand Off?

On 600 SQuAD items the 3.7B alone reaches F1 52.2 and gets 35.5% of items wrong ($F1 < 0.5$). A logistic probe on its pre-action hidden state (last prompt token, layers 12/18/24, nested 5-fold) predicts those failures with AUROC .749, better than its own semantic entropy (.689), sample disagreement (.683), max-probability (.582) or entropy (.584) (Table 14). But the 7B reading the passage reaches only 55.7, the mapped cache 51.8 and the verbal handoff 47.0, so no trigger makes a handoff pay at equal latency (Figure 7): the probe-triggered text handoff climbs from 52.2 to 55.7 as the handoff rate goes to 1, the cache handoff stays flat at 52, and only an oracle trigger gains (56.4 at rate .4). On this task the receiver is not enough better than the sender for a handoff of any kind to be worth it; the probe result stands, the policy result waits for a task with a real gap.

H. Additional Figures

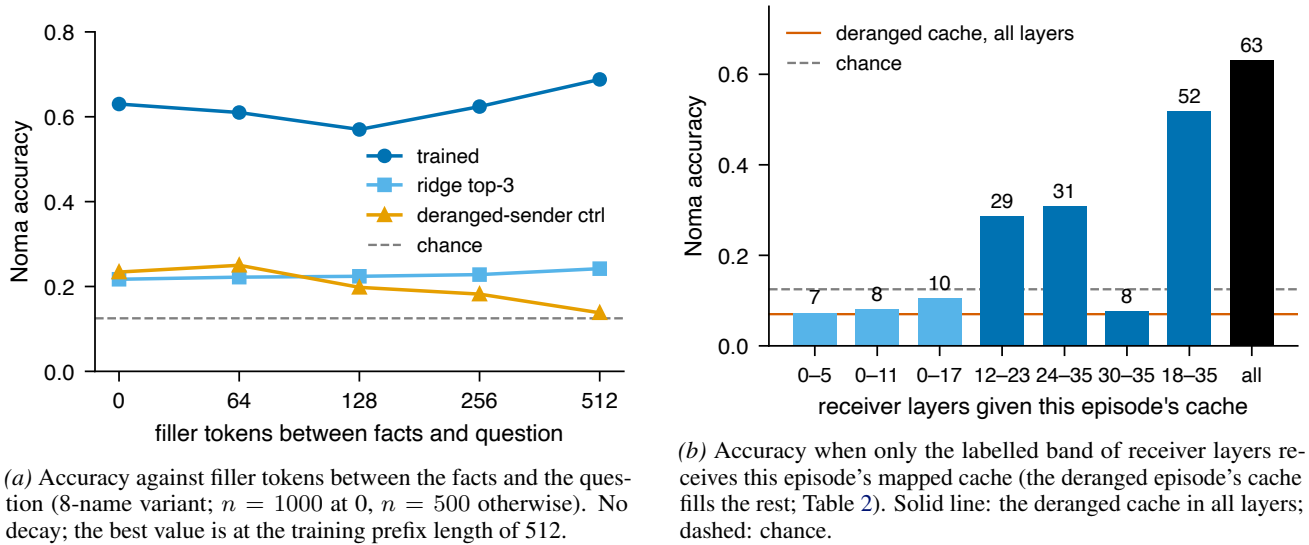


Figure 5. Distance sweep and layer-band injection.

Table 13. Every compression of the seed-2 projector’s mapped cache. KB per token is for 512-token SQuAD passages; low-rank truncation and mean-vector fill have a per-sequence overhead, so their per-token size depends on sequence length (for the 29-token Noma facts, rank 32 exceeds the full cache). Noma: 6-name variant, $n = 300$, percent; SQuAD: $n = 300$, token-F1. “filler” rows fill the untransmitted layers or heads with the receiver’s own cache of a neutral text of the same length.

spec	KB / token	Noma project	Noma derange	SQuAD F1
none (full, bf16)	144	75.3	2.3	53.0
int8	72	76.0	2.7	52.1
int4	36	75.0	3.7	51.6
rank 32	78.7	75.0	2.3	45.3
rank 16	39.3	75.0	2.3	29.6
rank 8	19.7	51.7	7.0	17.2
layers 18–35, zeros	72	24.3	9.7	8.3
layers 18–35, zeros, int8	36	24.7	9.7	7.8
4 heads, zeros	72	25.7	8.7	8.2
2 heads, zeros	36	22.3	9.7	5.9
layers 12–35, mean fill	96.4	35.0	11.0	28.8
layers 18–35, mean fill	72.7	33.7	11.7	21.4
layers 18–35, mean fill, int4	18.7	33.3	11.7	21.1
layers 24–35, mean fill	48.9	29.3	12.0	14.5
4 heads, mean fill	72.7	23.0	11.3	15.3
layers 12–35, filler	96	30.7	11.7	18.9
layers 18–35, filler	72	28.7	12.0	16.1
layers 18–35, filler, int4	18	30.0	11.7	16.9
layers 24–35, filler	48	25.3	10.0	13.3
4 heads, filler	72	26.7	10.0	12.8
projector bottleneck 32 (70 M params)	144	64.3	5.7	53.7
projector bottleneck 16 (62 M params)	144	61.3	7.7	51.6

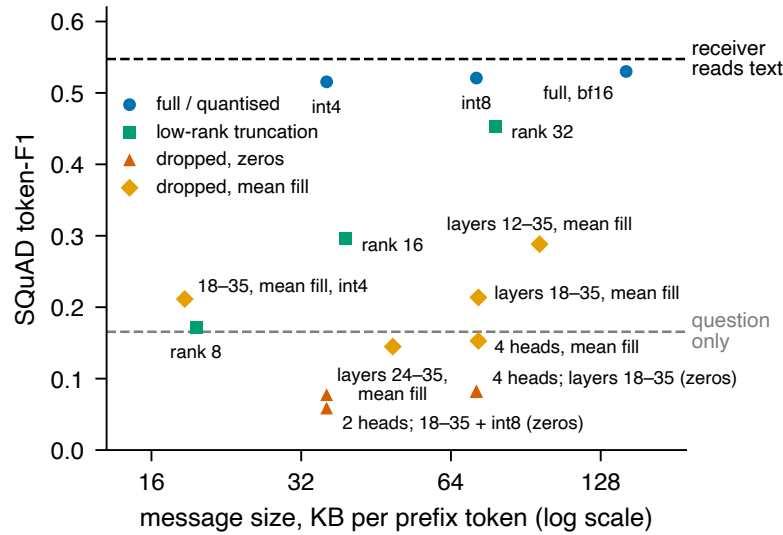


Figure 6. Message size against SQuAD F1 for every compression of the mapped cache in the zero- and mean-fill sweep ($n = 300$). Quantisation is free; low-rank truncation and dropping layers or heads are not. The two “zeros” points at 72 and 36 KB each hide two coincident specs.

Table 14. Handoff policy on 600 SQuAD items. Top: AUROC of each trigger at predicting the sender’s own failures ($F1 < 0.5$). Bottom: F1 of each arm when every item is handed off (rate 1).

sender (3.7B) alone: F1 52.2, wrong on 35.5% of items	
trigger	AUROC
probe on the sender’s hidden state	.749
semantic entropy (10 samples)	.689
sample disagreement	.683
max-probability	.582
entropy	.584
receiver (7B) sees	
F1 at rate 1	
the passage as text	55.7
the mapped cache	51.8
the sender’s answer + confidence word	47.0

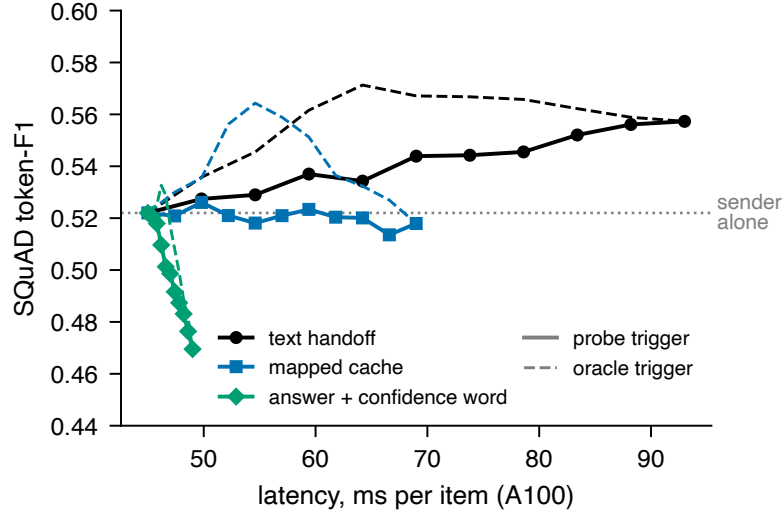


Figure 7. F1 against latency per item as the handoff rate goes from 0 (sender alone, 45 ms) to 1, for the three handoff arms under the probe trigger (solid) and an oracle trigger that hands off exactly the sender’s failures (dashed). 600 SQuAD items. Only the oracle gains; the probe-triggered text handoff climbs to the receiver’s own 55.7, the cache handoff stays flat.

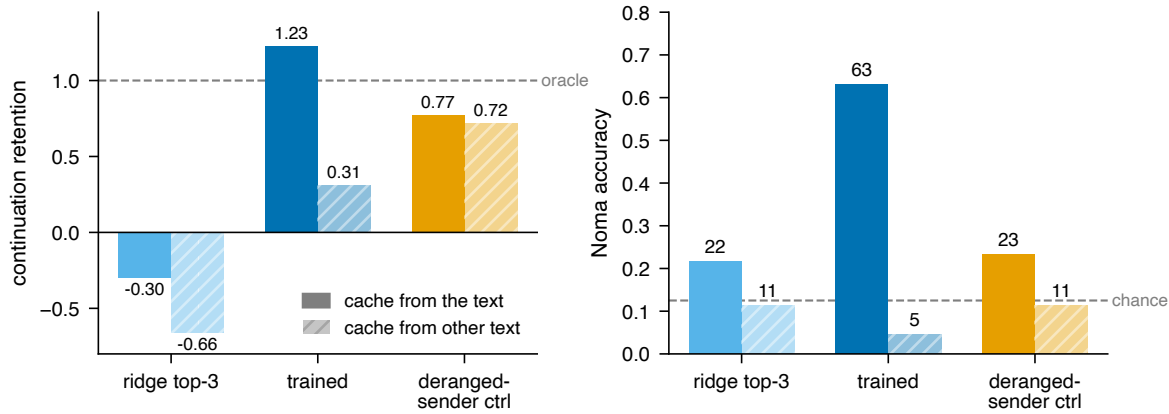


Figure 8. Retention and the binding test disagree about the deranged-sender control. Left: continuation retention on 128 fresh held-out sequences for the three projectors, with the sender reading the true prefix (solid) or a swapped one (hatched). Right: Noma accuracy for the same three projectors ($n = 1000$, 8-name variant), *project* arm (solid) and *derange* arm (hatched). Retention ranks the deranged-sender projector at 0.77, most of which survives a swapped sender; the binding test puts it at chance.

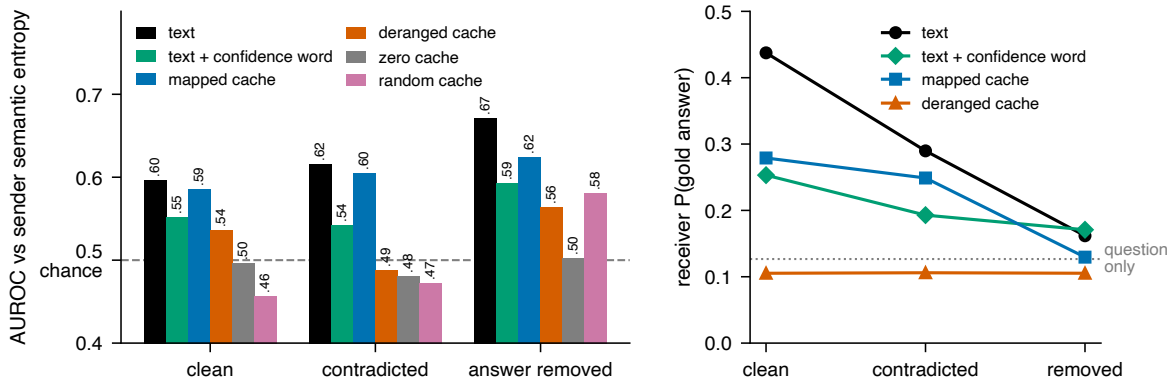


Figure 9. Left: AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, per handoff and passage variant (377 items), including the zero and moment-matched random caches. Right: the receiver’s $P(\text{gold})$ from clean to contradicted to removed passages. The cache path is less confident than the text path on clean passages and keeps the original answer ahead under a contradiction.